

HUJI Research Monitor

Healthcare · Weekly Digest · Week 30 · July 20, 2026

EXECUTIVE SUMMARY

This week's digest highlights a robust portfolio of innovations poised for commercial translation, spanning next-generation precision medicine, advanced therapeutic platforms, and transformative technologies addressing global challenges. Key themes include novel diagnostics and targeted therapies for oncology and neurological disorders, alongside breakthrough platforms in single-cell genomics, AI-driven drug discovery, and efficient genetic delivery. Additionally, we showcase advancements in high-precision medical devices, durable infection control solutions, and sustainable agricultural biocontrol agents, offering diverse opportunities for strategic investment and partnership.

#1

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Off-the-shelf stem cell therapy offers significant hope for remyelination in MS and other demyelinating disorders."

WHY NOW

Multiple Sclerosis represents a large market with an urgent unmet need for therapies that repair nerve damage. This scalable cell therapy addresses a critical bottleneck in restorative neurological treatment.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#2

CBFB mutations predict endocrine therapy benefit in estrogen receptor-positive breast cancer.

35/50

Main Researcher: Albert Grinshpun

[Diagnostics] [Genomics] [Clinical]

"Novel biomarker enables personalized endocrine therapy for ER+ breast cancer, improving patient outcomes."

WHY NOW

Companion diagnostics are essential for precision oncology, reducing ineffective treatments and healthcare costs. This test promises to better identify patients who will benefit most from widely used hormone therapies.

COMMERCIAL ANGLE

Development of a companion diagnostic test to personalize endocrine therapy selection for ER+ breast cancer patients, reducing ineffective treatments and improving clinical outcomes.

<https://pubmed.ncbi.nlm.nih.gov/42238456/>

#3

35/50

sc-rDSeq: a robust and cost-effective full-length total RNA sequencing method for single cells reveals multilayered heterogeneity in drug-resistant lung cancer cells.

Main Researcher: Oren Ram

[Genomics] [Diagnostics] [Drug Discovery] [Clinical]

"Breakthrough single-cell sequencing platform offers unprecedented resolution and cost-effectiveness for drug discovery."

WHY NOW

The exploding field of single-cell genomics demands more comprehensive and affordable tools to uncover disease mechanisms. This platform can accelerate precision diagnostics and therapeutic development.

COMMERCIAL ANGLE

The technology presents a high-value platform for next-generation precision diagnostics and drug discovery tools by providing more comprehensive transcriptomic data for identifying rare disease drivers.

<https://pubmed.ncbi.nlm.nih.gov/41978262/>

#4

34/50

Allele-specific antisense oligonucleotide treatment rescues

Main Researcher: Tamar Harel

[Drug Discovery] [Genomics] [Clinical]

"Precision allele-specific ASOs target genetic mutations, offering highly personalized therapy for rare disorders."

WHY NOW

The ability to neutralize pathogenic genetic variants without affecting healthy alleles marks a significant leap in gene therapy. This approach reduces side effects, opening new therapeutic avenues for challenging genetic diseases.

COMMERCIAL ANGLE

The ability to precision-target single-nucleotide mutations opens a high-value therapeutic pathway for rare genetic disorders and personalized genomic medicine.

<https://pubmed.ncbi.nlm.nih.gov/42239176/>

#5

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"Proprietary coating enhances hospital disinfection, combating multidrug-resistant bacteria in healthcare environments."

WHY NOW

Hospital-acquired infections, especially from MDR bacteria, pose a severe threat and incur massive costs globally. This scalable, durable solution offers an immediate enhancement to existing infection control protocols.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#6

Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake.

33/50

Main Researcher: Eylon Yavin

[Drug Discovery] [Genomics] [Synthetic Biology]

"Revolutionary "naked" genetic delivery system simplifies cellular uptake of therapeutic oligonucleotides."

WHY NOW

Delivery remains a major hurdle for genetic medicines, with current methods often complex and expensive. This elegant solution bypasses traditional barriers, potentially unlocking a new generation of cost-effective intracellular therapeutics.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

<https://pubmed.ncbi.nlm.nih.gov/42294022/>

#7

ASO Author Reflections: Robotic Cytorreduction at the Vascular Interface: Redefining Surgical Limits.

32/50

Main Researcher: Liron Kogan

[Medical Device] [Clinical] [Imaging]

"Robotic surgical system revolutionizes complex tumor removal, enhancing precision near critical vascular structures."

WHY NOW

The growing market for robotic surgery seeks innovations that expand treatability and reduce human error in challenging procedures. This system can redefine safety limits for oncological cytorreduction.

COMMERCIAL ANGLE

Development of a high-precision robotic surgical platform for oncological cytorreduction to reduce human error and expand the treatability of complex vascular-adjacent tumors.

<https://europepmc.org/article/MED/42120689>

#8

Universal rhythmic architecture uncovers two modes of neural dynamics.

28/50

Main Researcher: Ayelet N Landau

[Diagnostics] [Medical Device] [Software/AI] [Neuroscience]

"Universal neural rhythm architecture paves the way for advanced neurodiagnostics and brain-computer interfaces."

WHY NOW

Understanding brain activity is crucial for diagnosing and treating neurological conditions and developing next-gen human-machine interfaces. This fundamental discovery enables more precise and personalized neuro-technologies.

COMMERCIAL ANGLE

This universal architecture can be used to develop highly personalized neurodiagnostic software and advanced brain-computer interfaces that distinguish between stable cognitive states and rapid sensory responses.

<https://pubmed.ncbi.nlm.nih.gov/42218120/>

#9

Identifying maximally informative signal-aware representations of single-cell data using the information bottleneck.

28/50

Main Researcher: Mor Nitzan

[Drug Discovery] [Software/AI] [Genomics] [Immunology]

"AI framework distills complex single-cell data, accelerating drug discovery and precision medicine."

WHY NOW

The exponential growth of single-cell RNA sequencing data creates a critical need for advanced analytical tools. This AI framework efficiently identifies disease drivers and biomarkers, speeding up therapeutic development.

COMMERCIAL ANGLE

The framework can be commercialized as a high-throughput software platform for drug discovery and precision medicine to identify specific cellular biomarkers and disease drivers from noisy single-cell datasets.

<https://pubmed.ncbi.nlm.nih.gov/41928515/>

#10

RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

"Bacterial protein activates NK cells, offering a novel immunotherapy approach to combat cancer."

WHY NOW

Immunotherapy remains a frontier in cancer treatment, with strong interest in harnessing innate immune cells like NK cells. This discovery provides a unique mechanism for enhancing anti-tumor responses.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

<https://europepmc.org/article/MED/42065370>

#11

Paenibacillus dendritiformis C454 as a sustainable biocontrol agent for management of bacterial plant diseases.

25/50

Main Researcher: Yael Helman

[AgriTech] [Synthetic Biology]

"Sustainable biocontrol agent offers effective, eco-friendly solution for managing bacterial plant diseases."

WHY NOW

Global demand for sustainable agriculture is driving the shift away from chemical pesticides. This biological spray provides an environmentally sound and effective alternative for high-value crops.

COMMERCIAL ANGLE

Development of a sustainable, biological foliar spray to replace chemical pesticides in large-scale vegetable and fruit production.

<https://europepmc.org/article/MED/41975582>

#12

25/50

Nitric oxide inhibition ameliorates cortical proteomic changes in the Cntnap2^{-/-} and Shank3^{Δ4-22} mouse models of autism spectrum disorder.

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Proteomics] [Genomics]

"Validated pharmacological target for genetically-linked autism spectrum disorders promises first-in-class treatments."

WHY NOW

Autism spectrum disorders represent a significant unmet medical need, particularly for genetically defined subtypes. Identifying specific, actionable pathways paves the way for precision small molecule therapies.

COMMERCIAL ANGLE

The identification of specific nitric oxide signaling pathways provides a validated pharmacological target for developing first-in-class small molecule treatments for genetically-linked autism spectrum disorders.

<https://europepmc.org/article/MED/42010670>